# IBM's Prithvi foundation models for geospatial AI

**Prithvi is the most significant open-source geospatial foundation model family available today**, representing a landmark collaboration between IBM Research and NASA to apply the foundation model paradigm — proven in NLP — to Earth observation and weather/climate science. The family now spans three pillars: Prithvi-EO for satellite imagery understanding (Hugging Face) (100M–600M parameters), (ResearchGate) Prithvi-WxC for weather and climate modeling (2.3B parameters), (Hugging Face +2) and a tiny edge-deployable variant capable of running on satellites at 329 fps. (Maginative) (IBM) Pre-trained on NASA's petabyte-scale archives using self-supervised masked autoencoding, these models achieve state-of-the-art results on GEO-Bench (arXiv) while requiring dramatically less labeled data for downstream tasks (nasa) like flood mapping, wildfire detection, and crop classification. (arXiv +2) All models are released under permissive open-source licenses on Hugging Face, (arXiv +2) with the fine-tuning toolkit TerraTorch enabling configuration-driven adaptation to new tasks. (arXiv) (IBM)

## How a NASA–IBM partnership built a geospatial foundation model

The Prithvi project (Sanskrit for "Earth") (NASA Earthdata) emerged from NASA's Office of the Chief Science Data Officer, (NASA Science) which recognized that NASA's **100+ petabytes** of openly available Earth science data remained largely untapped by modern AI. (NASA Science) The collaboration brought together IBM Research's expertise in foundation model training with NASA's IMPACT (Interagency Implementation and Advanced Concepts Team) at Marshall Space Flight Center, (Clark University +2) led by Dr. Rahul Ramachandran, which provided domain expertise and data curation.

Development began in **January 2023**, with the first model — Prithvi-EO-1.0-100M — publicly released on Hugging Face in **August 2023**, described at the time as the largest open-source geospatial AI foundation model. (Clark University +2) The original research paper, "Foundation Models for Generalist Geospatial Artificial Intelligence" (arXiv:2310.18660), appeared in October 2023 (arXiv) with 32 co-authors spanning IBM, NASA, Clark University, and multiple universities. (Hugging Face) The collaboration expanded significantly for subsequent releases: Prithvi-WxC-1.0 (September 2024) was co-developed with Oak Ridge National Lab (Hugging Face)oratory, (Hugging Face) (IBM) while Prithvi-EO-2.0 (December 2024) involved (Hugging Face) **42 researchers from 12 institutions**, (IBM) including Germany's Jülich Supercomputing Centre, (Hugging Face) which provided the JUWELS supercomputer for training. (NASA Science +2) The project received the **AGU 2025 Open Science Recognition Prize** (IBM) for outstanding contributions to open science in Earth and space sciences. (Usra +2)

The motivating insight was straightforward: just as BERT and GPT demonstrated that self-supervised pre-training on massive unlabeled corpora produces transferable representations for language, the same approach could work for satellite imagery. (Medium) This would reduce dependence on expensive labeled datasets — a persistent bottleneck in remote sensing — and enable rapid fine-tuning for diverse downstream applications (IBM) from disaster response to agriculture monitoring. (arXiv +2)

## Architecture: Vision Transformers meet masked autoencoders in 3D

All Prithvi-EO models use a **Vision Transformer (ViT) backbone** with a **Masked Autoencoder (MAE)** self-supervised pre-training strategy, Hugging Face +3 extended from 2D images to 3D spatiotemporal cubes. GitHub Emergent Mind This architectural choice is central to understanding the model's capabilities.

**The MAE pre-training procedure** works as follows: input satellite image cubes of shape (C, T, H, W) — channels, time, height, width — are divided into non-overlapping 3D patches Hugging Face (tubelets) of size (t=1, h=16, w=16). Emergent Mind +2 For Prithvi-EO-2.0 with T=4 timestamps and 224×224 spatial resolution, this produces **784 tokens per sample** (4 × 14 × 14). During pre-training, **75% of these tokens are randomly masked**, and only the visible 25% pass through the transformer encoder Emergent Mind — a significant computational efficiency gain. A lightweight decoder then attempts to reconstruct the masked patches, trained with mean squared error (MSE) loss in pixel space. Emergent Mind +2 This forces the model to learn rich spatial, spectral, and temporal representations without any labeled data.

**Key architectural modifications** from standard 2D MAE/ViT distinguish Prithvi from conventional vision models:

- **3D patch embeddings** via 3D convolutional layers replace 2D patch projections, enabling native processing of multi-temporal satellite sequences GitHub +2
- **3D positional encodings** combine sinusoidal encodings along height, width, and time dimensions GitHub Hugging Face
- **Six spectral bands** (Blue, Green, Red, Narrow NIR, SWIR1, SWIR2) replace RGB, huggingface capturing information critical for vegetation indices, soil moisture estimation, and burn scar detection Medium Hugging Face
- **Temporal and Location (TL) embeddings** (in EO-2.0 TL variants) encode acquisition date (year + day-of-year) and geolocation (center latitude/longitude) Hugging Face using 2D sinusoidal encodings, added to tokens via learned weighted sums Hugging Face +2 — critical for resolving seasonal and geographic ambiguity in satellite imagery

The Prithvi-EO-1.0-100M uses a ViT-Base configuration: huggingface **768-dimensional embeddings, 12 encoder layers, 12 attention heads**, Hugging Face Hugging Face with an 8-layer decoder (512 dimensions, 16 heads). huggingface Prithvi-EO-2.0 scales this to ViT-Large (300M: 1024 dimensions, 24 layers, 16 heads) and ViT-Huge (600M: 1280 dimensions, 32 layers, 16 heads).

Prithvi-WxC-1.0 takes a fundamentally different architectural approach: a **2.3-billion-parameter encoder-decoder** Hugging Face using a scalable 2D Vision Transformer arXiv GitHub with **alternating local and global attention** inspired by Hiera, MaxViT, and SWIN transformers. Emergent Mind +3 Its pre-training objective is hybrid, combining masked reconstruction (50% masking) with explicit forecasting loss. arxiv +3 Crucially, it is **grid-free** — making no assumptions about the underlying data topology — enabling it to scale from global to regional resolution without architectural changes. IBM IBM The model has 25 encoder blocks and 5 decoder blocks, Hugging Face processes two input timestamps, and generates one (possibly future) output timestamp. Hugging Face

## Every Prithvi variant mapped out

The Prithvi family has grown substantially since 2023. Below is a complete catalog of all known variants as of early 2026.

### Prithvi-EO-1.0 (August 2023)

| Specification | Value |
|---|---|
| Parameters | 100M |
| Architecture | ViT-Base MAE |
| Input temporal frames | 3 |
| Spectral bands | 6 (Blue, Green, Red, NIR, SWIR1, SWIR2) |
| Spatial resolution | 30m |
| Training data coverage | Contiguous US only |
| Training samples | ~175,000 chips (>1 TB) |
| Pre-training source | HLS V2 L30 |

### Prithvi-EO-2.0 (December 2024)

| Variant | Parameters | Architecture | TL Embeddings | GPU-hours |
|---|---|---|---|---|
| EO-2.0-100M | 100M | ViT-Base | No | — |
| EO-2.0-300M | ~300M | ViT-Large | No | ~21,000 |
| EO-2.0-300M-TL | ~300M | ViT-Large | Yes | ~21,000 |
| EO-2.0-600M | ~600M | ViT-Huge | No | ~58,000 |
| EO-2.0-600M-TL | ~600M | ViT-Huge | Yes | ~58,000 |
| EO-2.0-tiny-TL | ~5M | Distilled | Yes | — |

All EO-2.0 variants use T=4 temporal frames, global training data (**4.2 million samples**, 2014–2023), (arXiv) (Clark University) and were trained on the JUWELS supercomputer (GitHub) at 400 epochs with cosine learning rate scheduling (max LR 5e-4, 40-epoch warmup). (arXiv +2) The 100M variant uses the same architecture as

EO-1.0 but benefits from the global dataset, yielding a **3% improvement** on GEO-Bench over EO-1.0. (arXiv) The tiny variant is **120× smaller** than the 600M model with less than 10% performance degradation, (IBM) designed for edge deployment on satellite computing platforms (tested at 329 fps on Unibap's iX10). (Maginative) (IBM)

### Prithvi-WxC-1.0 (September 2024)

| Specification | Value |
|---|---|
| Parameters | 2.3 billion |
| Architecture | Encoder-decoder ViT with local/global attention |
| Training data | MERRA-2 (160 variables, 1980–2019, 3-hourly) |
| Spatial resolution | 0.5° × 0.625° (global) |
| Pre-training objective | Mixed masked reconstruction + forecasting |
| Masking ratio | 50% |
| Encoder blocks | 25 |
| Decoder blocks | 5 |

Two pre-trained checkpoint flavors exist: prithvi.wxc.2300m.v1 (general use, variable lead times) and prithvi.wxc.rollout.2300m.v1 (Hugging Face) (optimized for autoregressive 6-hour rollout forecasting). (Hugging Face) (Hugging Face)

### Fine-tuned models

IBM and NASA have released numerous fine-tuned checkpoints: (Hugging Face) burn scar segmentation (EO-1.0 and EO-2.0), flood mapping via Sen1Floods11, multi-temporal crop classification, (GitHub) gravity wave parameterization (WxC), and weather downscaling. (Hugging Face) Enterprise-grade fine-tuned versions are branded as **IBM Granite Geospatial** models (Hugging Face) (e.g., granite-geospatial-wxc-downscaling, granite-geospatial-biomass, granite-geospatial-windforecasting). (IBM) (Hugging Face)

## Training data: from US satellite archives to global coverage

**Prithvi-EO models** are pre-trained on NASA's **Harmonized Landsat Sentinel-2 (HLS)** dataset (NASA Science) (Hugging Face) — a harmonized surface reflectance product combining Landsat 8/9 (USGS/NASA) and Sentinel-2A/2B (ESA) imagery at **30-meter spatial resolution** (nasa) with an average revisit period of 2–3 days. (PetaPixel) Of the 15 available bands, Prithvi uses the 6 common to both sensor families: Blue (B2), Green (B3), Red (B4), Narrow NIR (B8A), SWIR1 (B11), and SWIR2 (B12). (arXiv +3)

The training data scale increased dramatically between versions. EO-1.0 used approximately **175,000 multi-temporal chips** from the contiguous United States (Clark University) (Hugging Face) (>1 TB total). (Clark University) (Medium) EO-2.0 expanded to **4.2 million globally distributed time series samples** (arXiv) (IBM) plus 45,568 validation samples, (arXiv) spanning 2014–2023. (Clark University) Sampling was carefully stratified by land cover classes (using Copernicus Land Cover 100m) and ecoregions (RESOLVE Ecoregions dataset, **846 ecoregions** represented), with urban areas intentionally upsampled. (arXiv) Each sample consists of a temporal sequence of 4 timestamps with 1–6 month intervals between consecutive observations, (arXiv) captured as 256×256 pixel chips randomly cropped to 224×224 during training. (nasa) Strict cloud and missing-data filtering ensured data quality. (Emergent Mind) (arxiv)

**Prithvi-WxC** was pre-trained on an entirely different data domain: **NASA's MERRA-2 reanalysis dataset** (Modern-Era Retrospective Analysis for Research and Applications, Version 2). (Emergent Mind +5) The training set covers **40 years** (IBM) **(1980–2019)** at 3-hourly temporal resolution (OpenReview) and 0.5° × 0.625° spatial resolution globally. (arXiv) The model ingests **160 variables**: (Hugging Face) (arxiv) 20 surface variables, 10 variables across 14 vertical levels (140 level-variables), and 4 static variables (e.g., surface geopotential height, land fraction). Climatological baselines were computed from 2000–2019 to resolve the diurnal cycle, (OpenReview) and anomaly-based normalization was applied. Validation uses 2020–2023 data. Notably, fine-tuning for gravity wave parameterization uses ERA5 data rather than MERRA-2, demonstrating cross-reanalysis transfer capability. (arXiv) (arXiv)

## From flood maps to hurricane tracks: applications across Earth science

Prithvi's versatility across downstream tasks is perhaps its most compelling feature. The models have been fine-tuned and deployed across a wide spectrum of Earth observation and weather/climate applications.

**Disaster response** represents the most mature application domain. (Clark University) (NASA Science) For flood mapping using the Sen1Floods11 dataset (4,831 chips across 14 biomes and 6 continents), (Hugging Face) Prithvi-EO-2.0-600M with TL embeddings outperformed Prithvi-EO-1.0 by 5.6 points on IoU. (arXiv) (arXiv) A critical finding is Prithvi's **superior transferability**: while U-Net may outperform on in-sample data, Prithvi generalizes better to unseen geographies (Dagstuhl) (arXiv) — outperforming U-Net by ~4% mIoU on out-of-distribution test sets (Emergent Mind) (e.g., Bolivia). (Emergent Mind) (Emergent Mind) In a real-world deployment for the **October 2024 Valencia, Spain floods**, Prithvi mapped (IBM) approximately 120 km² of flood damage (IBM) (Medium) with **30% improved accuracy** versus traditional methods (Medium) by combining Sentinel-1 radar and Sentinel-2 optical data. (IBM) For burn scar detection, Prithvi-EO-2.0-600M-TL achieved **83.62 mIoU** on HLS Burns in the PANGAEA benchmark — the highest score among all evaluated models.

**Agricultural monitoring** leverages the multi-temporal capabilities. (NASA Science) Fine-tuned on USDA Crop Data Layer labels, (NASA Science +2) Prithvi-EO-2.0-600M achieved **~70.4% mIoU** for multi-temporal crop segmentation, (arXiv) and showed remarkable data efficiency — performance dropped only to 67.1% with 50% less training data. (arXiv) On the European Sen4Map dataset (335,125 time series), (arXiv) Prithvi-EO-2.0 (arXiv) ViT-H achieved **76.10% on land cover** and **84.60% on crop type** classification — top reported results. (arXiv) (TheCVF)

**Weather and climate applications** through Prithvi-WxC are equally impressive. For hurricane track forecasting (Category 3–5 Atlantic hurricanes, 2017–2023), Prithvi-WxC achieved a mean track error of **63.9 km** for Hurricane Ida — compared to 201.9 km for MERRA-2-trained FourCastNet and 262.3 km for ERA5-trained FourCastNet. ResearchGate arXiv For climate downscaling of MERRA-2 2m temperature, the model achieved spatial RMSE of **0.73–0.76 K**, IBM a >4× improvement over interpolation baselines. arXiv The gravity wave parameterization application — the first use of any foundation model for climate model parameterization NASA — converged **40% faster** than training from scratch, Hugging Face with final loss of 0.106 versus 0.17. arXiv

Additional documented applications include IBM above-ground biomass estimation Clark University in the Bolivian Amazon (2016–2024 deforestation monitoring), IBM urban heat island mapping IBM (Baltimore, summer 2023), IBM cloud gap imputation outperforming GAN baselines, Clark University +2 locust breeding ground prediction, and even elephant detection from drone imagery in an IBM–WWF partnership. IBM IBM

## Benchmark performance reveals strengths and honest limitations

On the **GEO-Bench** evaluation suite (12 datasets: 6 classification, 6 segmentation), Prithvi-EO-2.0-600M-TL achieved an average score of **75.6%** — an 8% improvement over Prithvi-EO-1.0 IBM and the best overall combined performance across all 12 datasets. IBM arXiv It outperformed six other geospatial foundation models NASA Technical Reports Serve… (MOCO, DINO, DeCUR, ScaleMAE, DOFA, Satlas) arXiv and performed well even on high-resolution tasks (0.1m aerial imagery) despite being pre-trained at 30m resolution. arXiv ResearchGate For classification tasks specifically, DOFA ranked first with Prithvi-EO-2.0-600M-TL second; for segmentation, Prithvi topped the leaderboard. arXiv

Independent benchmarks provide a more nuanced picture. arXiv On the **PANGAEA benchmark** (11 datasets), Prithvi excelled on HLS Burns (83.62 mIoU, highest) and Sen1Floods11 (90.37 mIoU, near-highest), but underperformed on multi-temporal tasks like PASTIS-R and DynamicEarthNet compared to uni-temporal models like CROMA. arXiv The **GEO-Bench-2** evaluation (2025, 19 datasets) found that no single model dominates across all capability groups — EO-specific models like Prithvi, TerraMind, and Clay are superior in multi-spectral and multi-temporal settings, while models pre-trained on natural images lead on high-resolution RGB tasks. IBM Emergent Mind

A key differentiator for Prithvi-WxC relative to weather AI models like GraphCast, Pangu-Weather, and FuXi: those are **single-task forecast emulators**, while WxC is a true foundation model supporting forecasting, downscaling, reconstruction, parameterization, and extreme event estimation from a single pre-trained backbone. arXiv +2

From an efficiency standpoint, Prithvi's continual pre-training required only **93.3 GPU-hours** compared to SatMAE's 768 GPU-hours (with correspondingly lower carbon footprint: 13.3 kg $CO_2$ versus 109.44 kg $CO_2$). Emergent Mind The full 600M model training consumed ~58,000 GPU-hours on 240 GPUs of the JUWELS supercomputer. arXiv nasa

## Open-source ecosystem: from Hugging Face to satellite hardware

All Prithvi models are hosted on Hugging Face under the ibm-nasa-geospatial organization Hugging Face (https://huggingface.co/ibm-nasa-geospatial), with **no authentication or gating required**. EO models use the **Apache 2.0 license**; IBM WxC uses **CDLA-Permissive-2.0**. Hugging Face All models are in PyTorch format. The collection has accumulated over **600,000 downloads** and **350+ citations** since initial release. IBM

The primary fine-tuning framework is **TerraTorch** GitHub (https://github.com/IBM/terratorch), an open-source toolkit GitHub IBM built on PyTorch Lightning and TorchGeo, IBM installable via pip install terratorch or conda-forge. arXiv GitHub TerraTorch supports configuration-driven fine-tuning GitHub via YAML files IBM with no code changes required, arXiv and natively supports Prithvi, Granite, TerraMind, Clay, SatMAE, Satlas, DOFA, and any model from timm or SMP. IBM GitHub Since version 1.1, it includes parameter-efficient fine-tuning (PEFT) support for LoRA, ViT-Adapter, and Visual Prompt Tuning (VPT). IBM

Key GitHub repositories include NASA-IMPACT/Prithvi-EO-2.0 (main EO-2.0 code with fine-tuning examples), GitHub NASA-IMPACT/hls-foundation-os (original EO-1.0 code using MMSegmentation), GitHub and NASA-IMPACT/Prithvi-WxC (weather/climate model). GitHub Prithvi-EO-2.0 was notably the **first non-text-input/non-text-output model onboarded to vLLM**, IBM enabling scalable serving through vLLM's pooling endpoint with GeoTIFF input/output processing. IBM IBM's **Geospatial Studio** provides a cloud-native interface guiding users through data curation, fine-tuning via TerraTorch, model deployment to watsonx.ai, and visualization. IBM IBM Enterprise-ready fine-tuned models are available as IBM Granite Geospatial models. Hugging Face IBM

For edge deployment, the tiny variant (~5M parameters, ~129 MB) has been tested on Unibap's iX5 and iX10 satellite computing platforms Maginative IBM — including the iX5 currently used on NASA satellites — achieving **329 fps** on 6-band 224×224 images. IBM The architecture uses a frozen encoder with uploadable task-specific decoder heads (~1–2 MB each), enabling mid-orbit model updates on satellites. Maginative IBM

## Key research publications for further reading

The three foundational papers define the Prithvi family. **Jakubik et al. (2023)**, "Foundation Models for Generalist Geospatial Artificial Intelligence" (arXiv:2310.18660), Hugging Face introduces Prithvi-EO-1.0 and establishes the architectural framework. arXiv Hugging Face **Szwarcman et al. (2024)**, "Prithvi-EO-2.0: A Versatile Multi-Temporal Foundation Model for Earth Observation Applications" (arXiv:2412.02732), arXiv presents the scaled-up global models with TL embeddings and comprehensive benchmarking. arXiv +2 **Schmude et al. (2024)**, "Prithvi WxC: Foundation Model for Weather and Climate" (arXiv:2409.13598), Hugging Face details the 2.3B-parameter weather/climate model and its multi-task capabilities. arxiv +3

Supporting papers include arXiv "Seeing Through the Clouds: Cloud Gap Imputation with Prithvi Foundation Model" arXiv (arXiv:2404.19609), "Geospatial foundation models for image analysis: evaluating and enhancing NASA-IBM Prithvi's domain adaptability" arXiv (arXiv:2409.00489), arXiv "Assessment of a new GeoAI foundation model for flood inundation mapping" ResearchGate (arXiv:2309.14500), and "TerraTorch:

The Geospatial Foundation Models Toolkit" (arXiv:2503.20563). arXiv The models have been presented at AGU 2024/2025, IEEE GRSS workshops, ML4EO 2025, GitHub and CVPR 2025.

## Conclusion

Prithvi represents a paradigm shift in how geospatial AI research is conducted — moving from task-specific models trained from scratch to adaptable foundation models that transfer knowledge across applications, geographies, and even data domains (MERRA-2 pre-training transferring to ERA5 fine-tuning). arXiv Three insights stand out for researchers entering this space. First, **data diversity matters more than data volume**: the 3% improvement from simply retraining the same 100M architecture on global versus US-only data demonstrates the value of geographic breadth. ResearchGate arXiv Second, **temporal and location metadata are underutilized signals**: the TL embedding variants consistently outperform their non-TL counterparts, arXiv suggesting that encoding acquisition context improves representation quality even when downstream tasks lack this metadata. ResearchGate Third, the honest finding from independent benchmarks like PANGAEA and GEO-Bench-2 that **no single foundation model dominates all tasks** IBM Emergent Mind should guide practical decisions — Prithvi excels in multi-spectral, multi-temporal settings and cross-geographic transfer, while other architectures may be preferable for high-resolution RGB tasks Emergent Mind or specific multi-temporal benchmarks. For graduate researchers, the fully open ecosystem — Apache-licensed models, TerraTorch fine-tuning, NASA Technical Reports Serve… vLLM serving, and extensive Colab notebooks — makes Prithvi an exceptionally accessible entry point for foundation model research in Earth observation. arXiv